

Textbook EIA Work Progress Logging

Navit Dori

April 15, 2025

1 Uploading data from OECD and Statcan and Integrating Them

1.1 File *GDP_comp.py*

In *GDP_comp.py* I upload OECD data and Statcan data and integrate them by

1. Converting Statcan from CAD to USD
2. Mapping the Statcan sectors (more detailed) to OECD sectors

GDP_comp.py is the same as *func_upload_data.py* where I upload OECD and statcan data into *income_multipliers*.

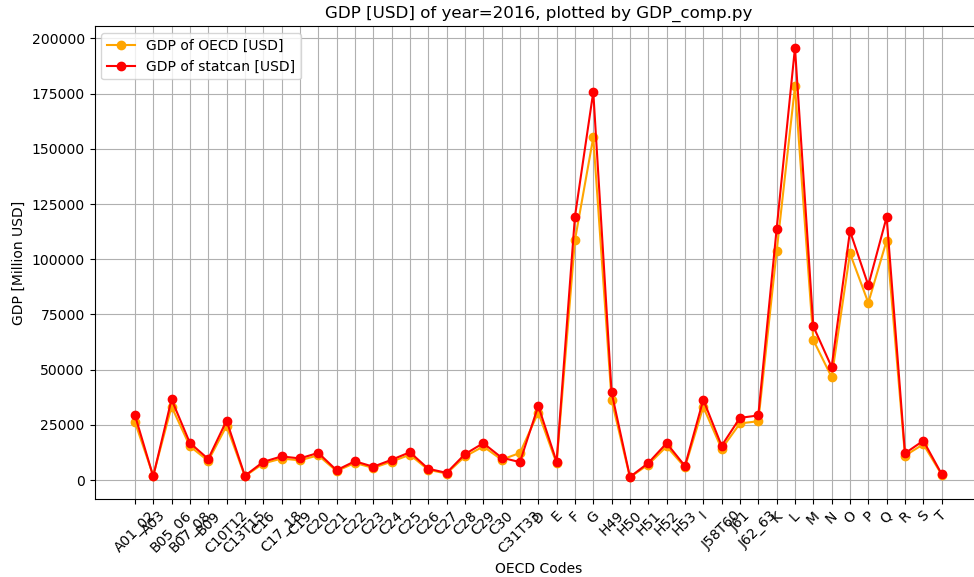


Figure 1: Value Added by Sector (StatCan, 2016)

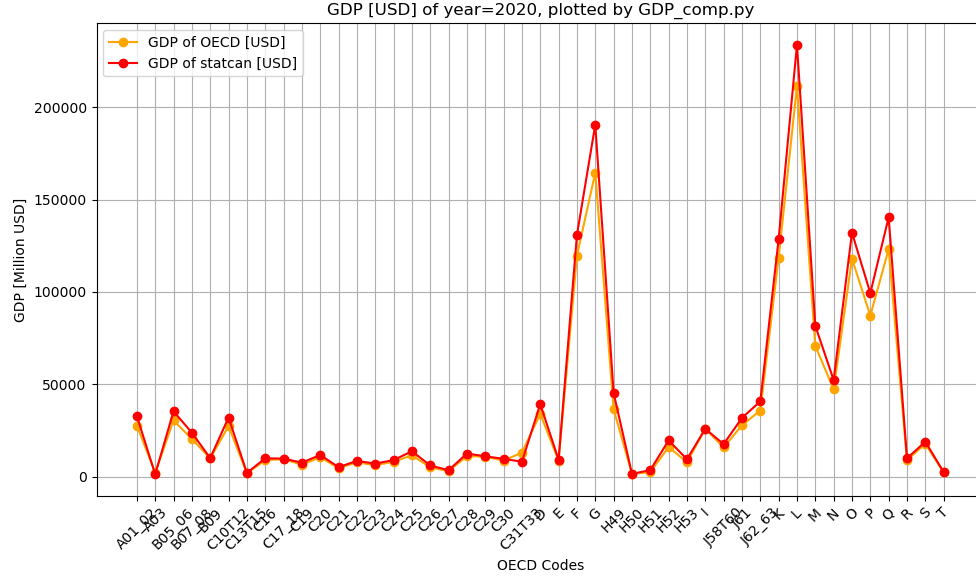


Figure 2: Value Added by Sector (StatCan, 2020)

1.2 Plotting T as function of year

plot_T_vs_time.py plots T of the 6 ICT sectors as function of year. Sector G (wholesaling) has variability in T .

1.3 Comparing OECD and statcan: GDP, II and output

Using *OECD_statcan_comp.py* comparing GDP and II and output between OECD and statcan.

1.4 Uploading all data from statcan - not just GDP

In *read_statcan.py* I upload all data from statcan - not just GDP. I need the “compensation of employees” for the closed model. I updated *func_data_upload.py* to *func_data_upload2.py* to include all data of statcan.

1.5 Output prediction

In *output_multipliers_all_sectors.py* I plot the predicted output versus the real output. The simple model gives very exact results.

Simple model: the predictions are accurate and consistent (see Fig. 5, 6) even in 2020 which shows the biggest divergence of T (graph not shown).

The prediction of the closed model is not very accurate. Possible reasons include:

1. A mistake in the implementation of the model;
2. Inaccuracies in the conversion from CAD to USD (PPP uploaded versus PPP experimental);

Column Values in T per Sector, by Year

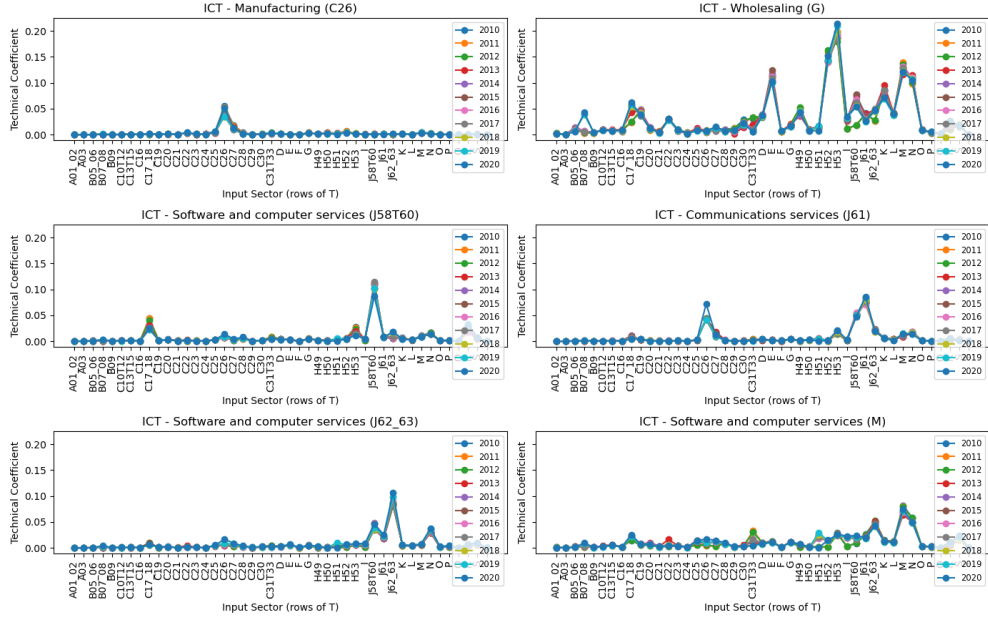


Figure 3: Value Added by Sector (StatCan, 2010-2020)

3. Errors introduced by the unification of OECD and StatCan data that are caused by the different data collection methods.

1.6 Summary of files

upload_data.py - this is uploading only GDP from statcan, and not a function. The early version of *func_upload_data2.py* *output_multipliers_all_sectors.py* where I calculate multipliers and plot predictions *OECD_statcan_comp.py* comparing GDP and II and output between OECD and statcan.

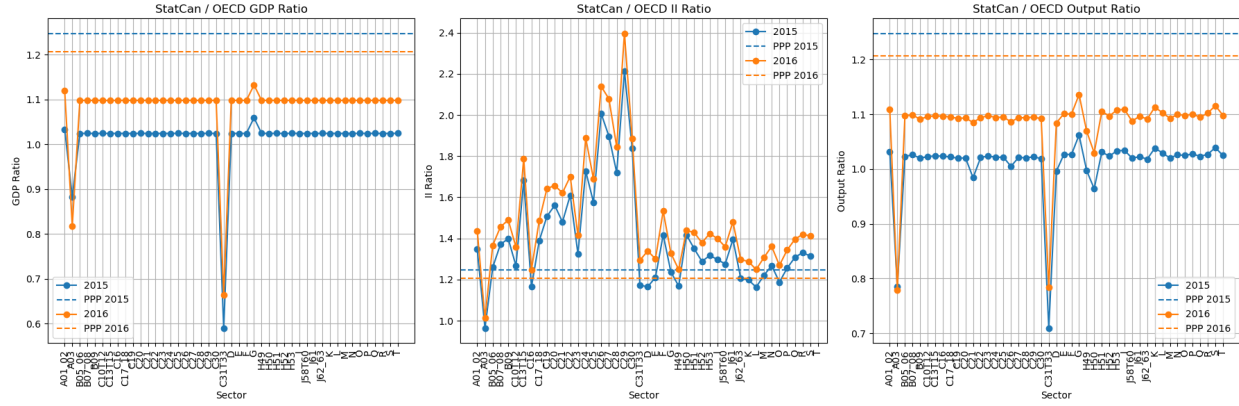
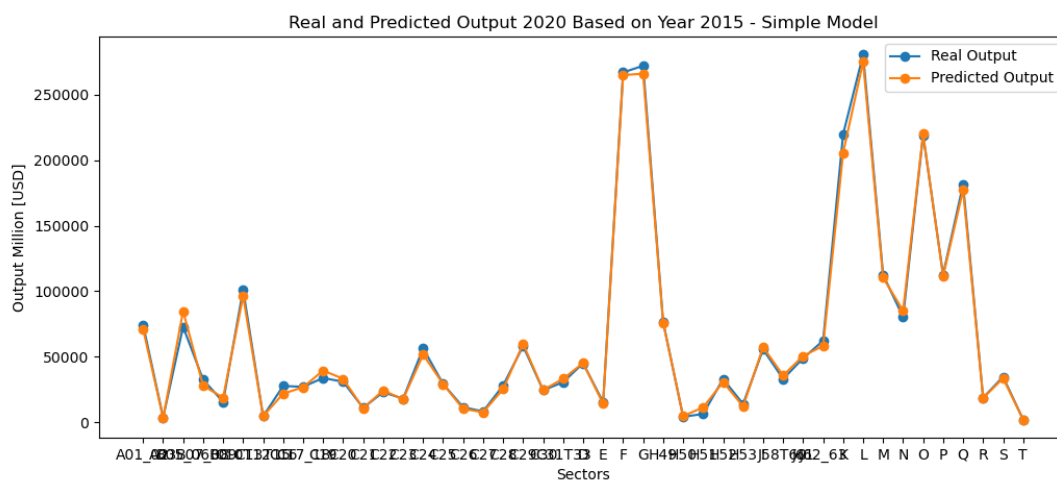
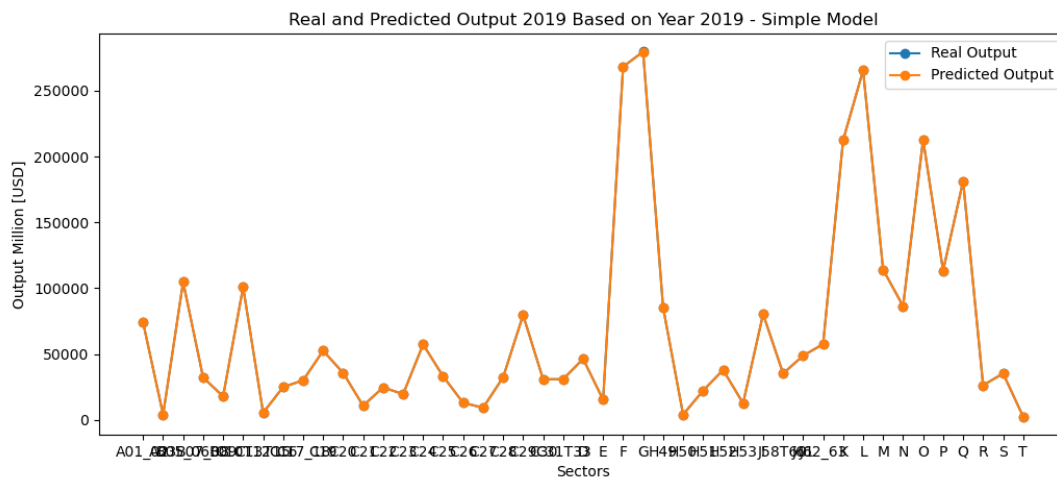


Figure 4: Plots of the ratio Statcan/OECD (CAD/USD) of Output, Intermediate Input and GDP. Also plotted are PPP, the CAD to USD conversion ratios uploaded form Statcan. It is clear that the actual ratios of output and GDP, which should reflect the PPP are far from these theoretic values. Also clear that the Output and GDP are generally similar, except for the sectors A03, C31T33, G, H50.

2 Appendix A: Sector Details

Statcan Code	Description	OECD code
A	Agriculture, forestry and fishing	-
A01	Crop and animal production, hunting and related service activities	
A02	Forestry and logging	A01_02
A03	Fishing and aquaculture	A03
B	Mining and quarrying	-
B05	Mining of coal and lignite	
B06	Extraction of crude petroleum and natural gas	B05_06
B07	Mining of metal ores	
B08	Other mining and quarrying	B07_08
B09	Mining support service activities	B09
C	Manufacturing	-
C10	Manufacture of food products	
C11	Manufacture of beverages	
C12	Manufacture of tobacco products	
C10T12	Manufacture of food products; beverages and tobacco products	C10T12
C13	Manufacture of textiles	
C14	Manufacture of wearing apparel	
C15	Manufacture of leather and related products	C13_15
C16	Manufacture of wood and of products of wood and cork, except furniture	
C17	Manufacture of paper and paper products	
C18	Printing and reproduction of recorded media	C16_18
C19	Manufacture of coke and refined petroleum products	C19
C20	Manufacture of chemicals and chemical products	C20
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	C21
C22	Manufacture of rubber and plastics products	C22
C23	Manufacture of other non-metallic mineral products	C23
C24	Manufacture of basic metals	C24



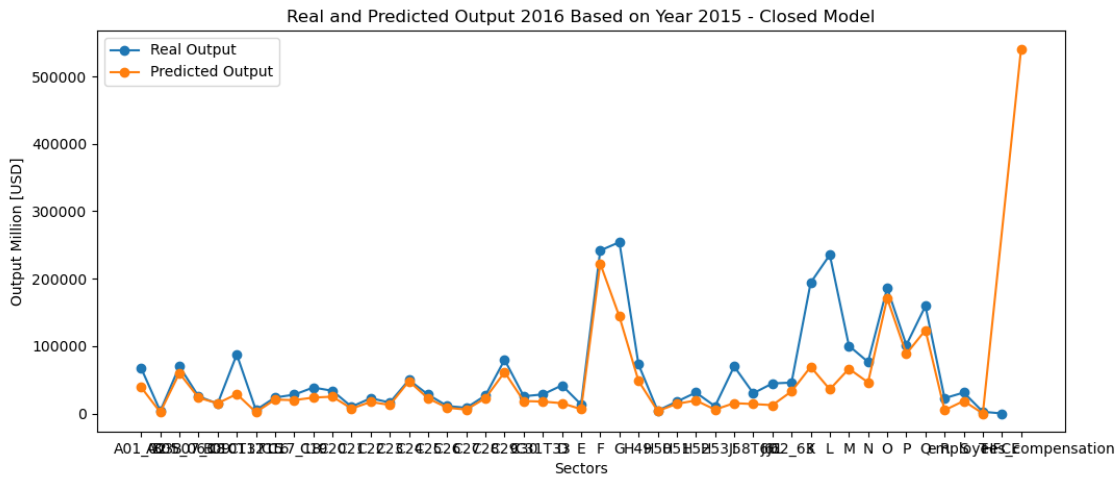


Figure 7: Closed model with PPP uploaded from statcan. Plotted by *output_multipliers_all_sectors.py*

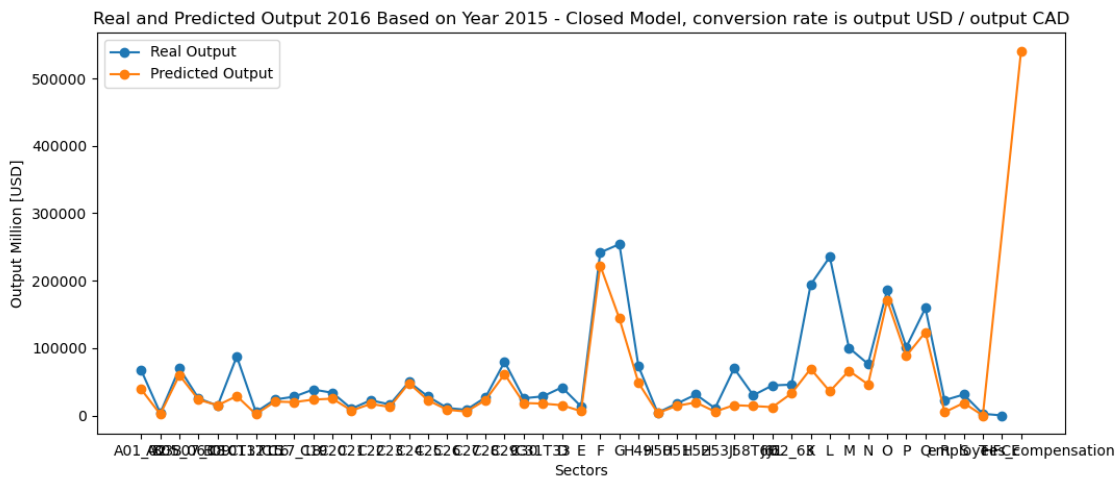


Figure 8: Closed model with experimental PPP - a rough estimation taken from the comparison of OECD GDP and statcan GDP. Plotted by *output_multipliers_all_sectors.py*